

EEE4114F Class Test 2

June 2021

Duration: 1 hours

Total: 40 marks

Empl ID:

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Instructions:

- This is a closed-book test but you may use a calculator.
- There are two parts to this test.
- **Part A** has *two* questions totalling 20 marks. You must answer all of them.
- **Part B** has *two* questions totalling 20 marks. You must answer all of them.
- Part A and part B each count 50% of the final mark.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 1 hour.

Part A:	Question:	1	2	Total
	Points:	10	10	20
	Score:			

Part B:	Question:	3	4	Total
	Points:	10	10	20
	Score:			

Part A: Digital Signal Processing

Question 1 [10 marks]

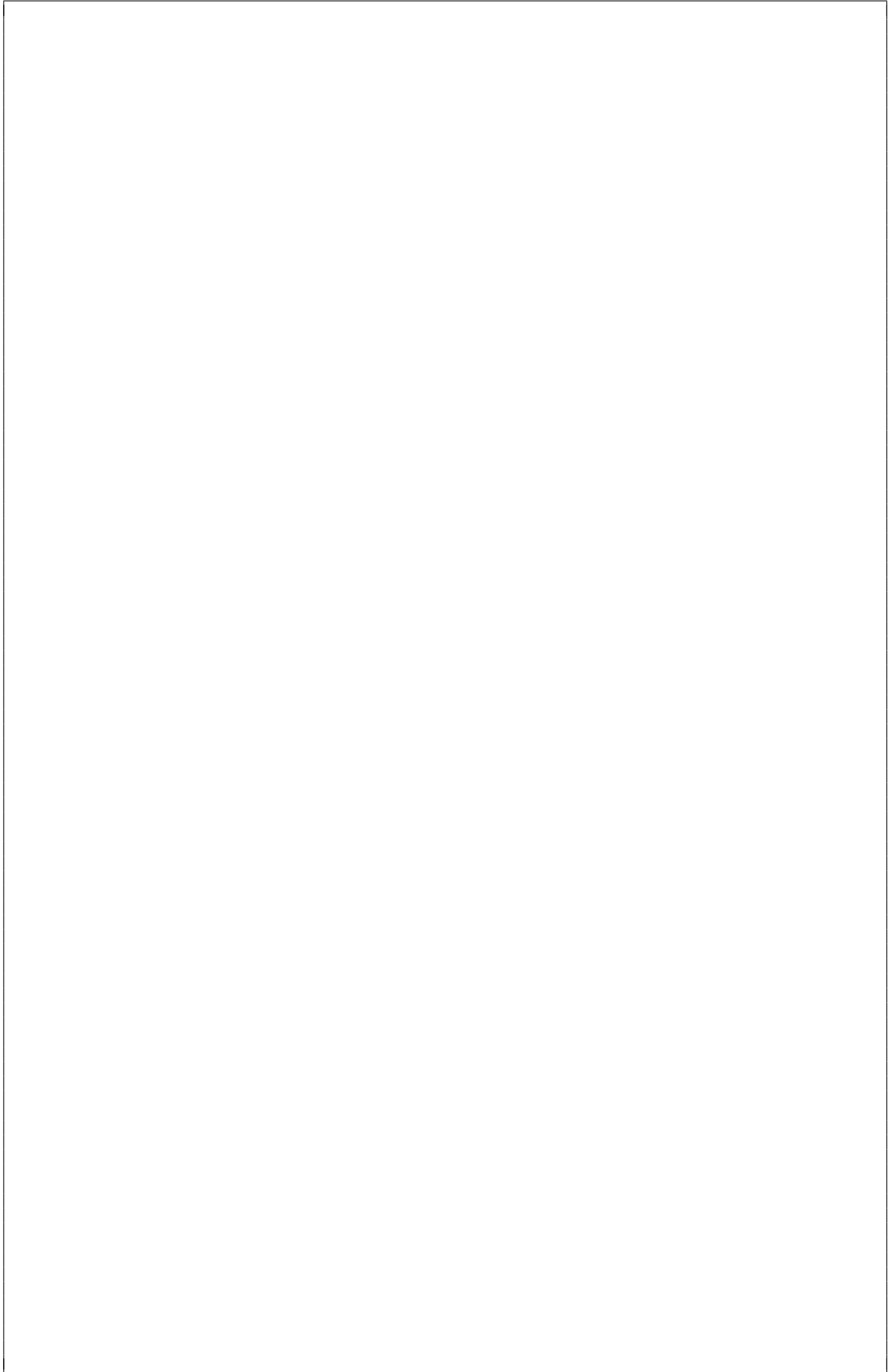
The continuous-time signal

$$x_c(t) = \frac{\sin(4000\pi t)}{4000\pi t}$$

is sampled with a sampling period T to obtain the discrete-time signal

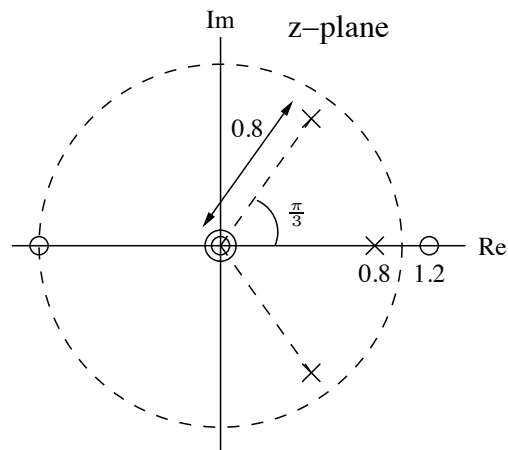
$$x[n] = \frac{\sin(\pi n/3)}{\pi n/3}.$$

- Determine a choice for T consistent with this information.
- Is your choice for T above unique? If so, explain why. If not, specify another choice of T consistent with the information given.
- Specify and sketch $X(e^{j\omega})$ over the range $0 \leq \omega \leq 4\pi$.

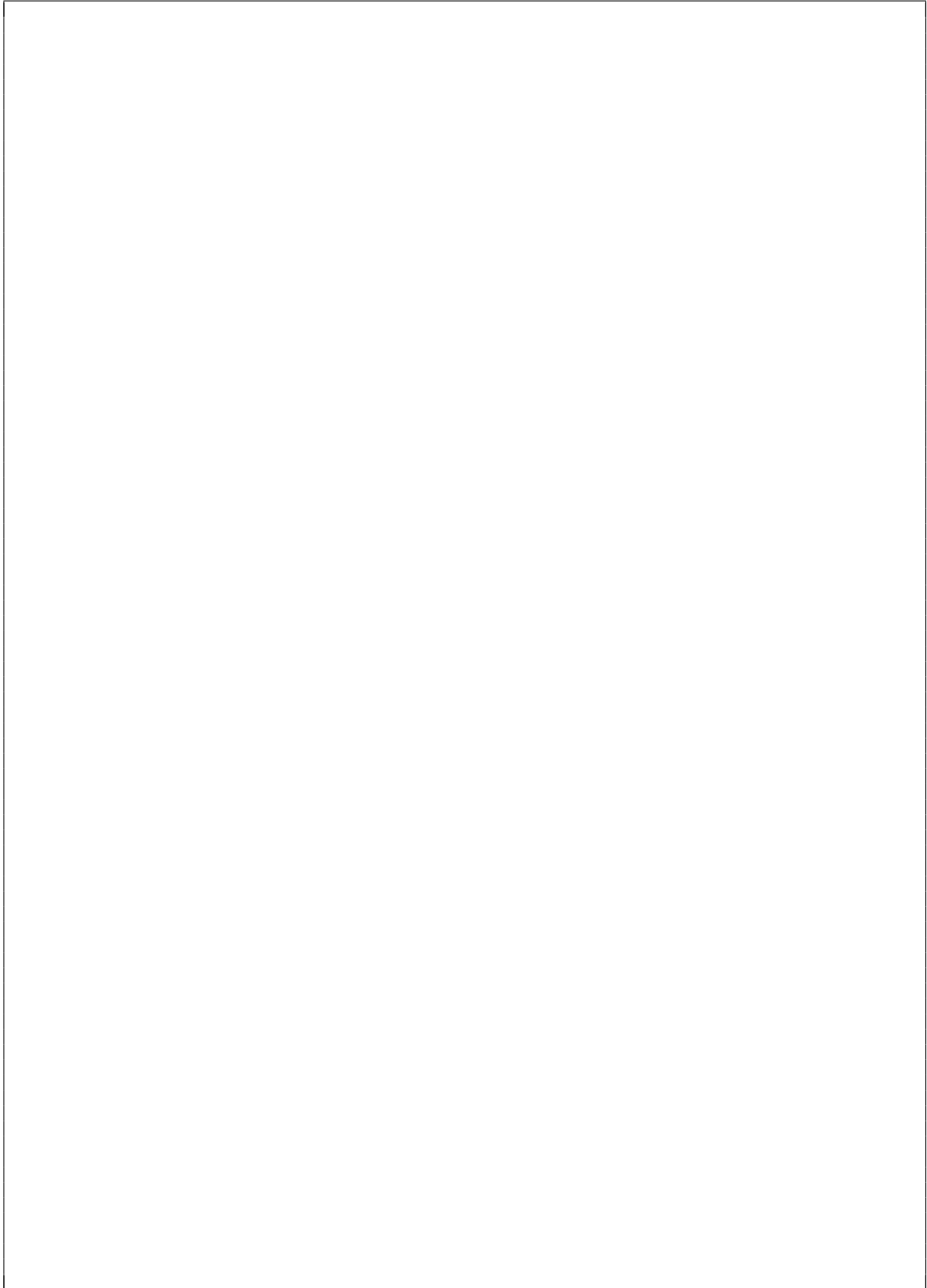


Question 2 [10 marks]

Consider an LTI system with the following pole-zero configuration:



- Give an expression for the system function $H(z)$.
- Sketch the magnitude response of the system.
- What type of filter does this system represent?
- What is the approximate phase response of the system at $\omega = 0$?



END OF PART A

Part B: Machine Learning

Question 3 [10 marks]

Suppose you are developing a self-driving car that has access to a forward-facing camera.

You train three different models: A, B, and C, as part of a task to identify street signs from the camera. You train and evaluate the models using 5-fold cross-validation with results shown as a box plot in Figure 1.

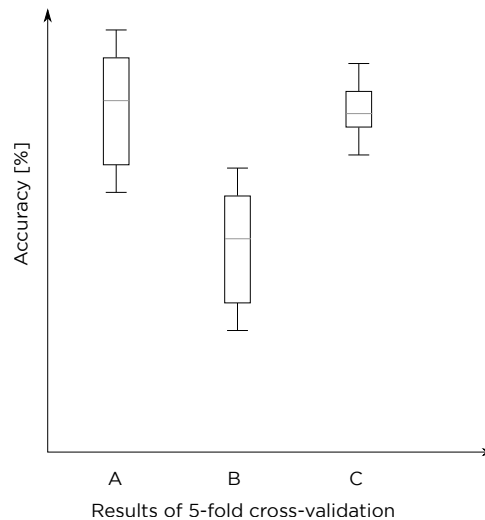


Figure 1: Box plots for three trained models using 5-fold cross-validation

- Explain what k-fold cross-validation is and why it is useful for evaluating a model's ability to generalise to unseen data compared to training and evaluating with a single holdout split. (2)
- Figure 2 shows a plot of model capacity vs error. Which points on this plot (points indicated as 1, 2, or 3 on the x-axis) would you expect to correspond to each model (A, B, and C)? Explain your decisions. (6)

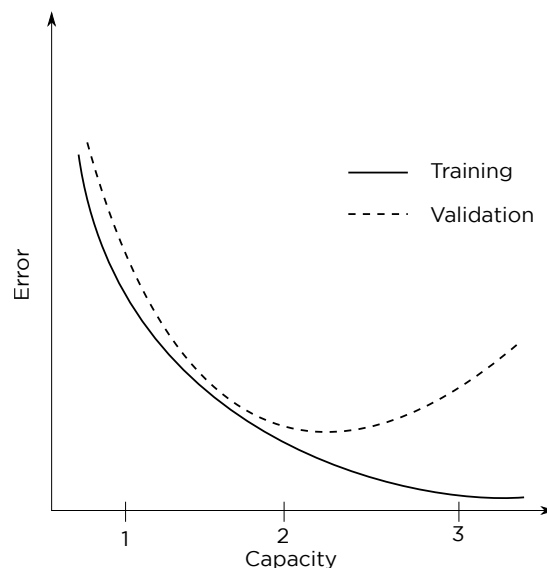
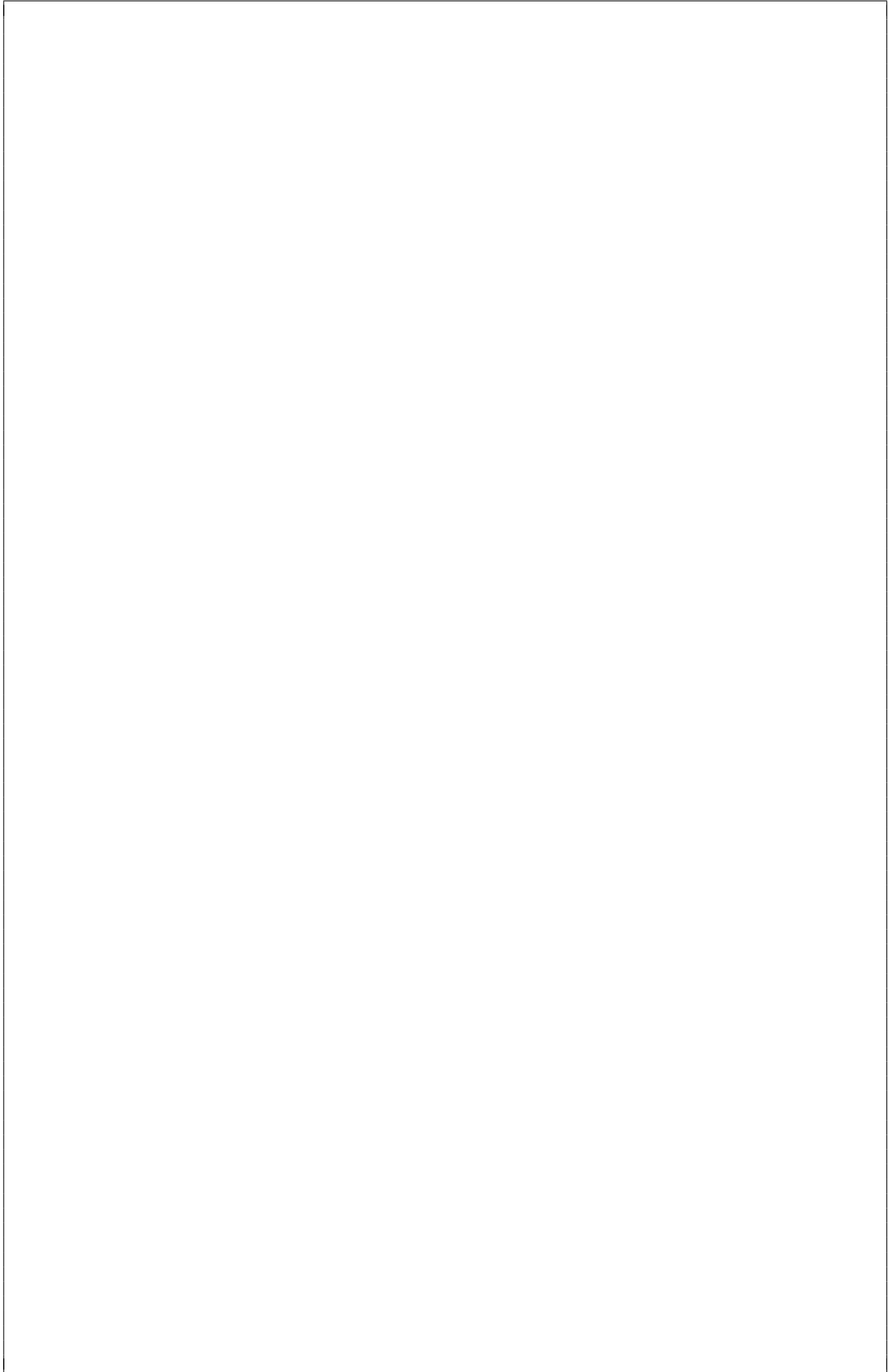
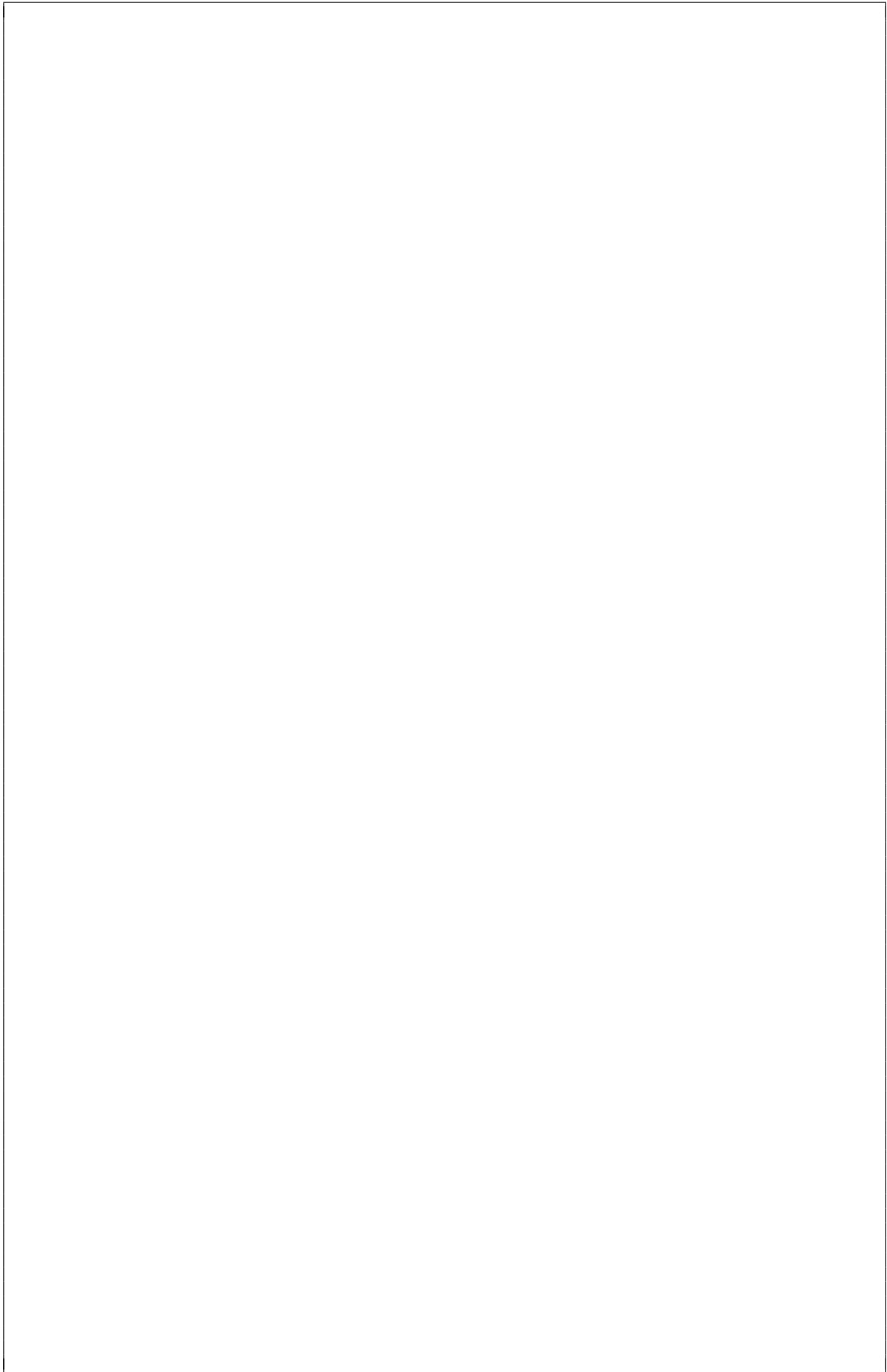


Figure 2: Model capacity vs. error

- Which model would you choose if you wanted the real-world performance to be reliable and close to the predicted performance. Explain your answer. (2)





Question 4 [10 marks]

Suppose you need to develop a system that measures distance to an obstacle and predicts a safe speed to travel at. You collect a data set, $\mathcal{D}$, from a professional driver consisting of input-output pairs (x_i, y_i) where x_i is the distance to an obstacle measured in m , and y_i is the target variable, which in this case is the speed of the vehicle measured in m/s .

$$\mathcal{D} = \{(1, 4.8), (3, 15.2), (1.5, 7.3), (2, 10.2)\}$$

You then decide to fit a linear regression model of the form $h_{\theta}(x) = \theta_0 + \theta_1 x$ to it using gradient descent optimization methods. The model's parameters are randomly initialised such that

$$\theta_0 = 0.1, \theta_1 = 0.3$$

. The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{N} \sum_{i=1}^N (h_{\theta}(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

Suppose you decide to divide the data set into two, equally sized minibatches, keeping the order as it is such that

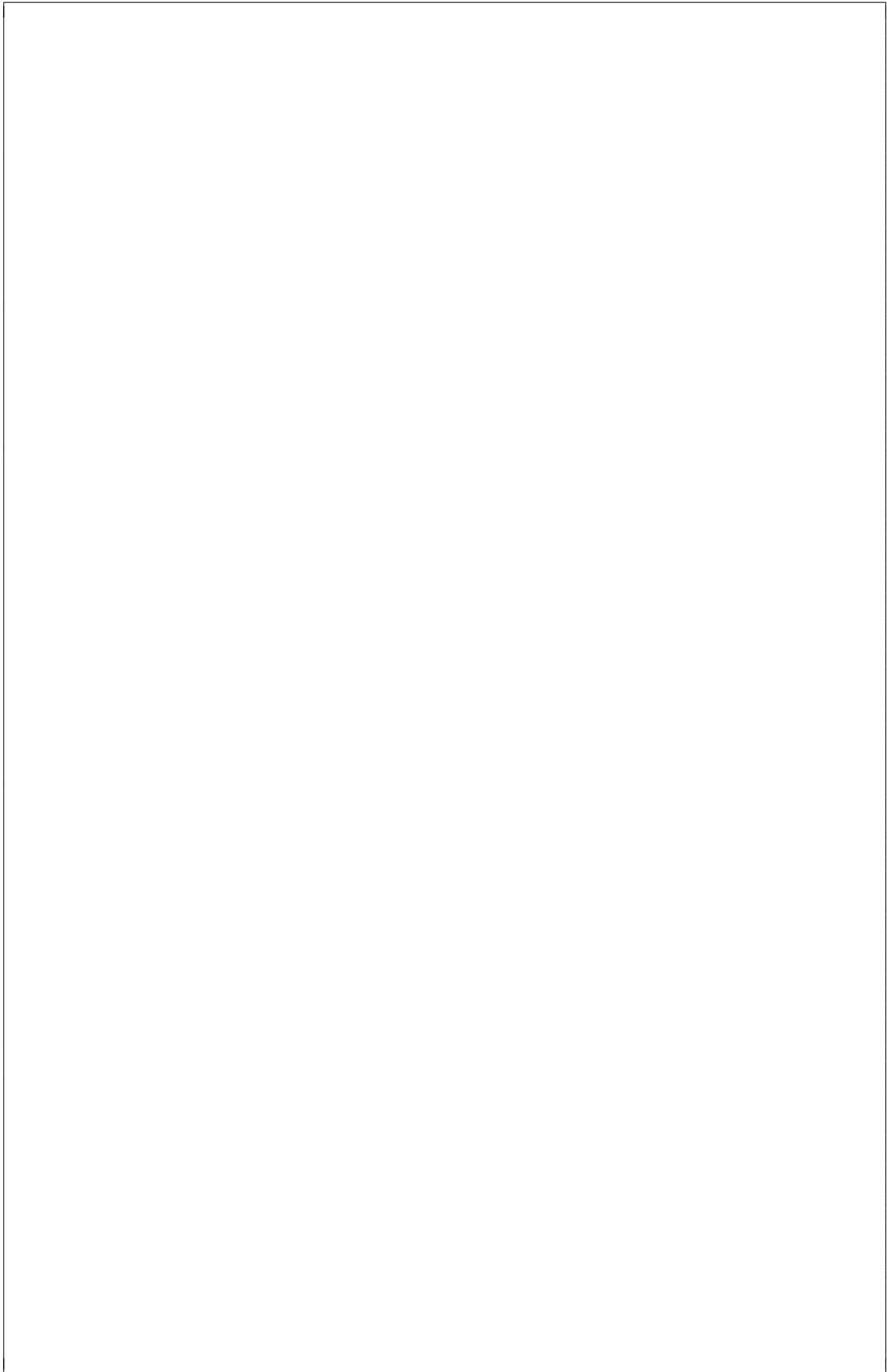
$$\mathcal{D} = \{((1, 4.8), (3, 15.2)), ((1.5, 7.3), (2, 10.2))\}$$

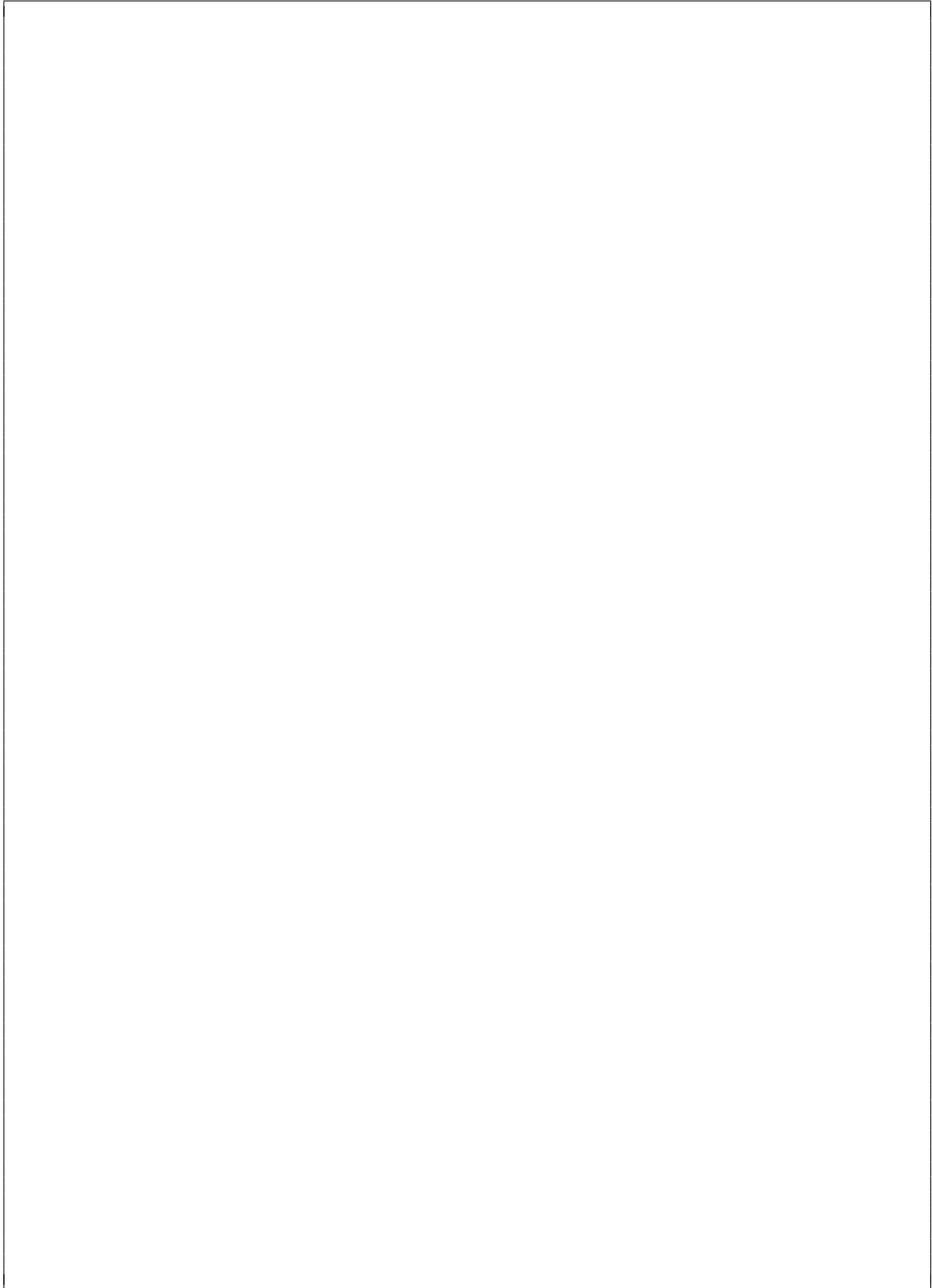
A single training iteration uses one minibatch and the model parameters are updated after **each** iteration. The update rule is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.001$. Answer the following questions

- Write down the expressions for the gradients of the objective function with respect to the model's parameters $\frac{\partial J}{\partial \theta_0}$ and $\frac{\partial J}{\partial \theta_1}$. (2)
- Calculate the gradients of the parameters after one training iteration using the first minibatch. (2)
- Calculate the **model** parameters after the second training iteration using the second minibatch. (4)
- Your friend tells you increase the capacity of you model by adding more input features including: the colour of the obstacle, the day of the week, and the ambient temperature. Would it make sense for this application to use l_1 or l_2 regularization terms? Explain your answer. (2)





END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$